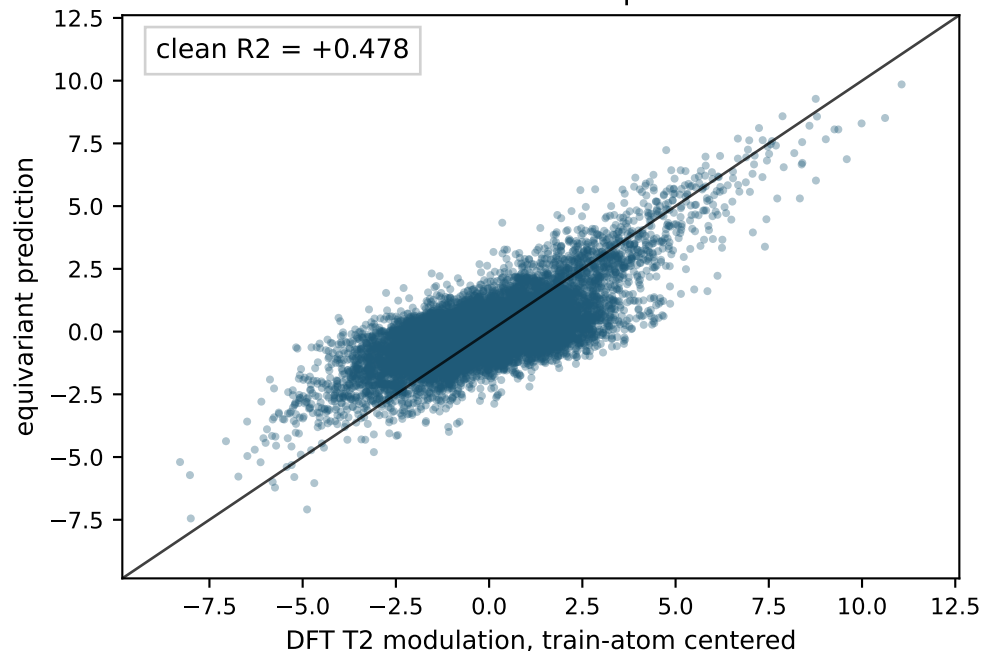
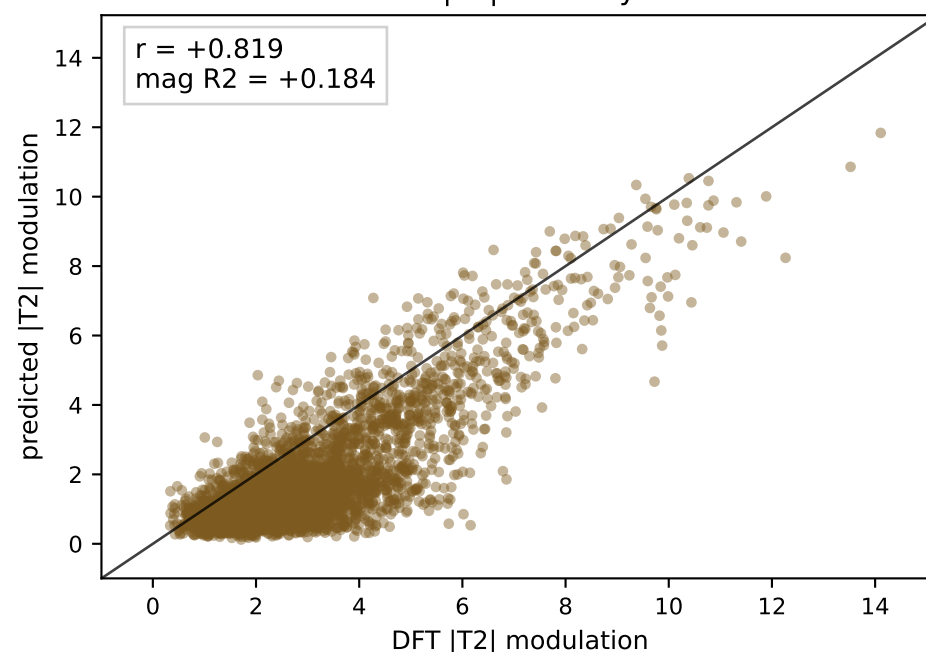


Equivariant shielding-tensor prediction - 1P9J

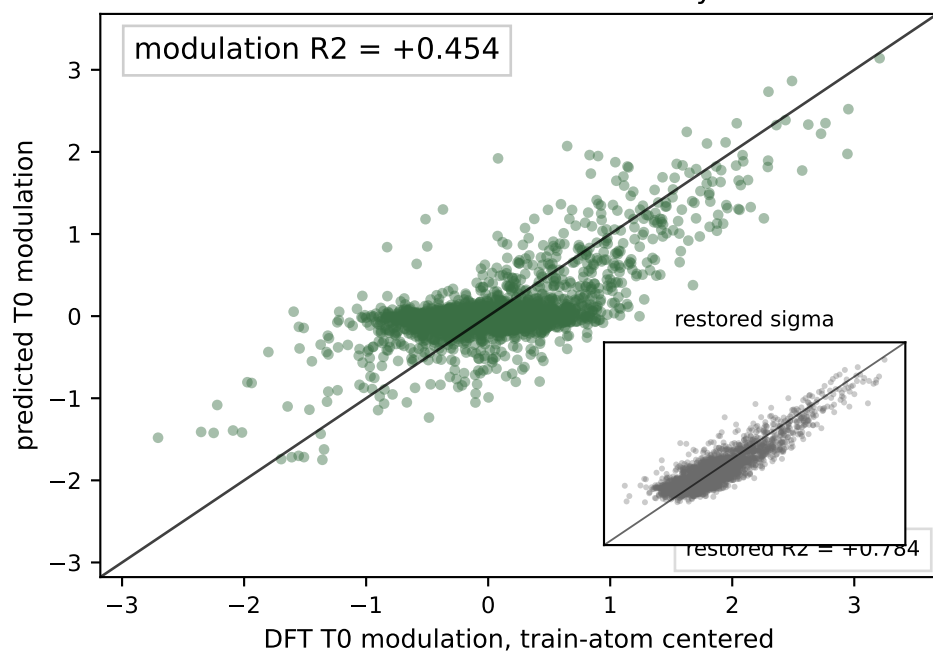
A. Held-out T2 components



B. $|T_2|$ recovery



C. T0 modulation recovery



D. Capability

Software tool: ingest a protein's local geometry and predict its NMR shielding tensor equivariantly on held-out frames. The tensor target is the 5-component T2 field, with a companion T0 channel.

Practical reading: for 1P9J, the tool trained on a subset of frames and predicted the rest of the same protein. Held-out T2 component R2 = +0.478; $|T_2|$ $r = +0.819$; T0 modulation R2 = +0.454.

Per-protein bootstrap: DFT a subset of a protein's frames, run this toolchain, and predict the rest of that protein at about 50% T2 variance recovery. Not great; better than nothing; real.

Run scale: 41 atoms, 500 frames, 20,500 atom-frame groups (16,318 train, 4,100 held out), learnable cross term.

Footnote: within-protein result; transferability/BMRB validation is separate.